Reactions of Amino Acids

Learning objectives

At the end of this presentation, students will be able to understand:

1. Enlist the types of reaction involve in amino acids synthesis and utilization.
2. Reactions of amino acids due to Amino group, Carboxyl group and Side chain of amino acids.
3. Their significance in relation to different biochemical processes.

Reactions of amino acids

1) Reactions due to amino group
2) Reactions due to carboxyl group
3) Reactions due to both amino and carboxyl groups
4) Reaction due to side chain

OR

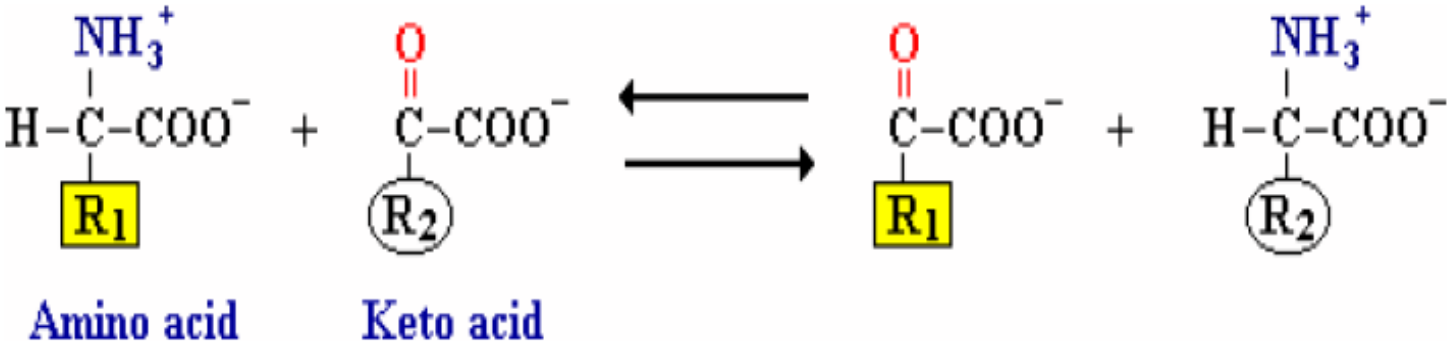
Another way to Categorized Reactions of Amino acids

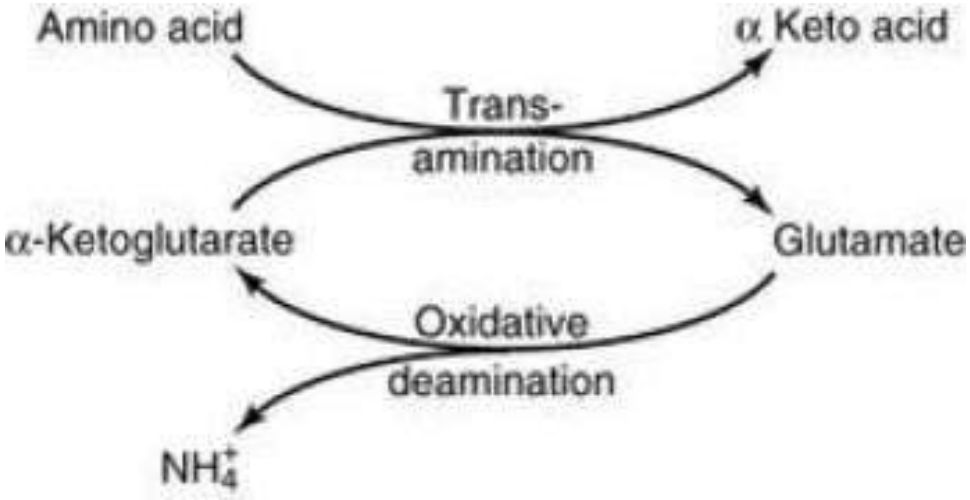
Reactions are enlisted as:

1. Transamination
2. Trans-methylation
3. Trans-peptidation
4. Deamination
5. De-amidation
6. Decarboxylation
7. Interconversion of amino acids

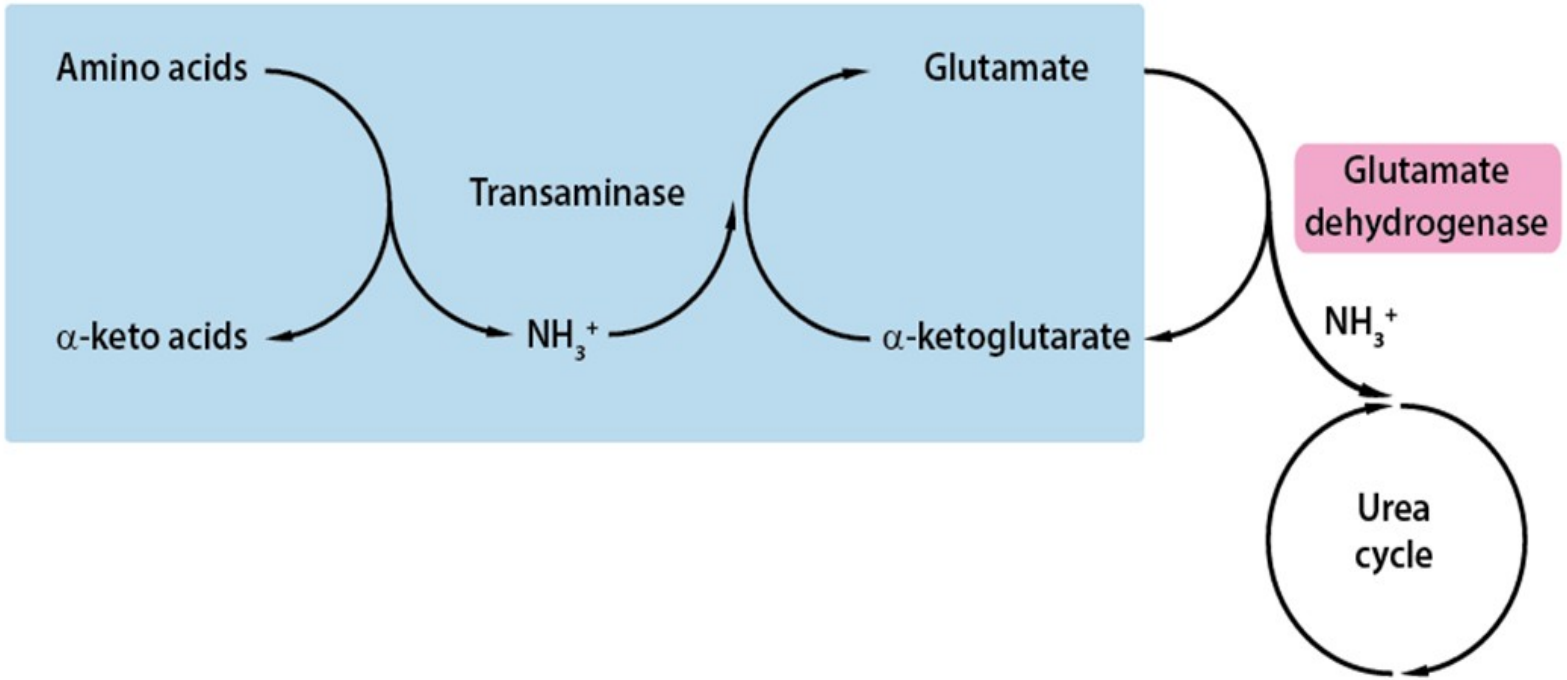
Reactions due to amino group (contd.)

ii) Transamination-Transfer of an α amino group from an amino acid to an α keto acid to form a new amino acid and a corresponding keto acid.





Transamination



Enzymatic Transamination

- Typically, α-ketoglutarate accepts amino groups
- L-Glutamine acts as a temporary storage of nitrogen
- L-Glutamine can donate the amino group when needed for amino acid biosynthesis
- All aminotransferases rely on the pyridoxal phosphate cofactor

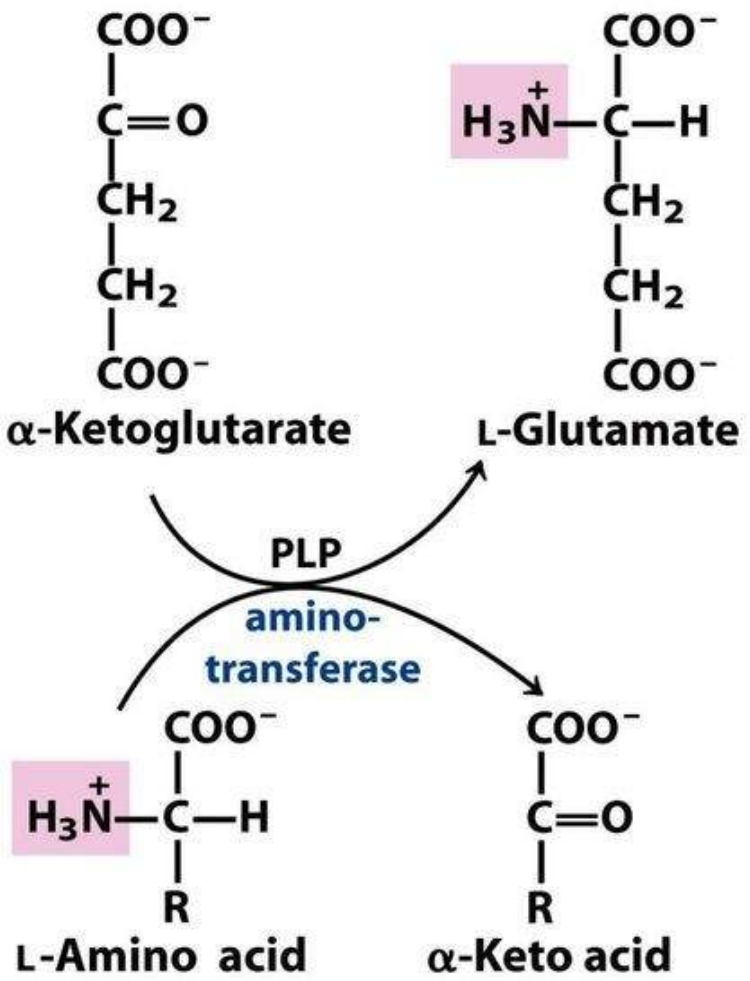


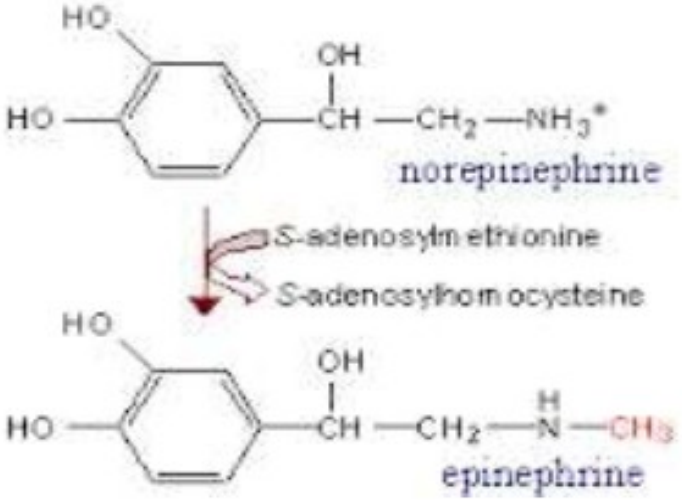
Figure 18-4
Lehninger Principles of Biochemistry, Fifth Edition
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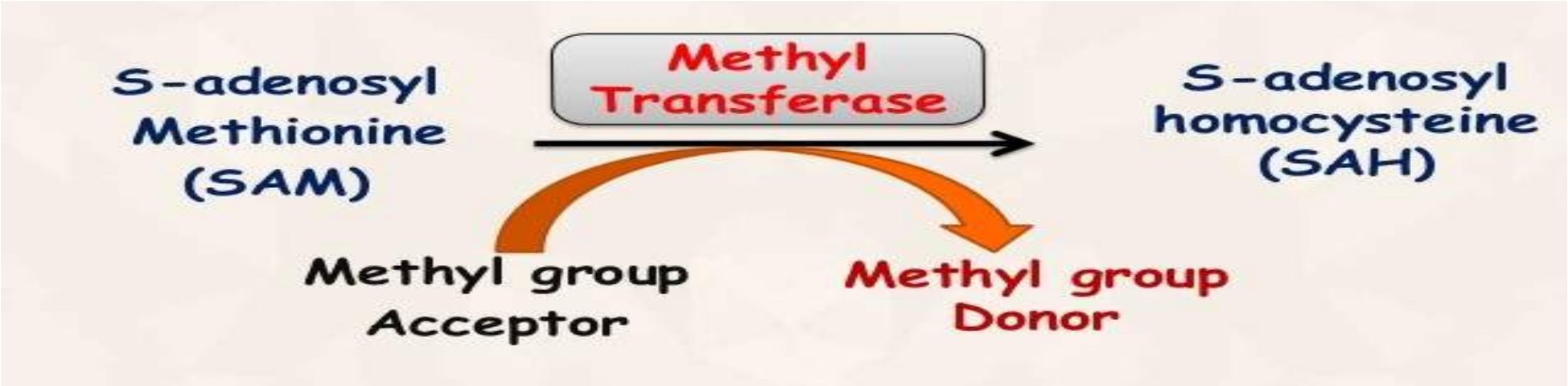
Reactions due to side chains (contd.)

3)Transmethylation

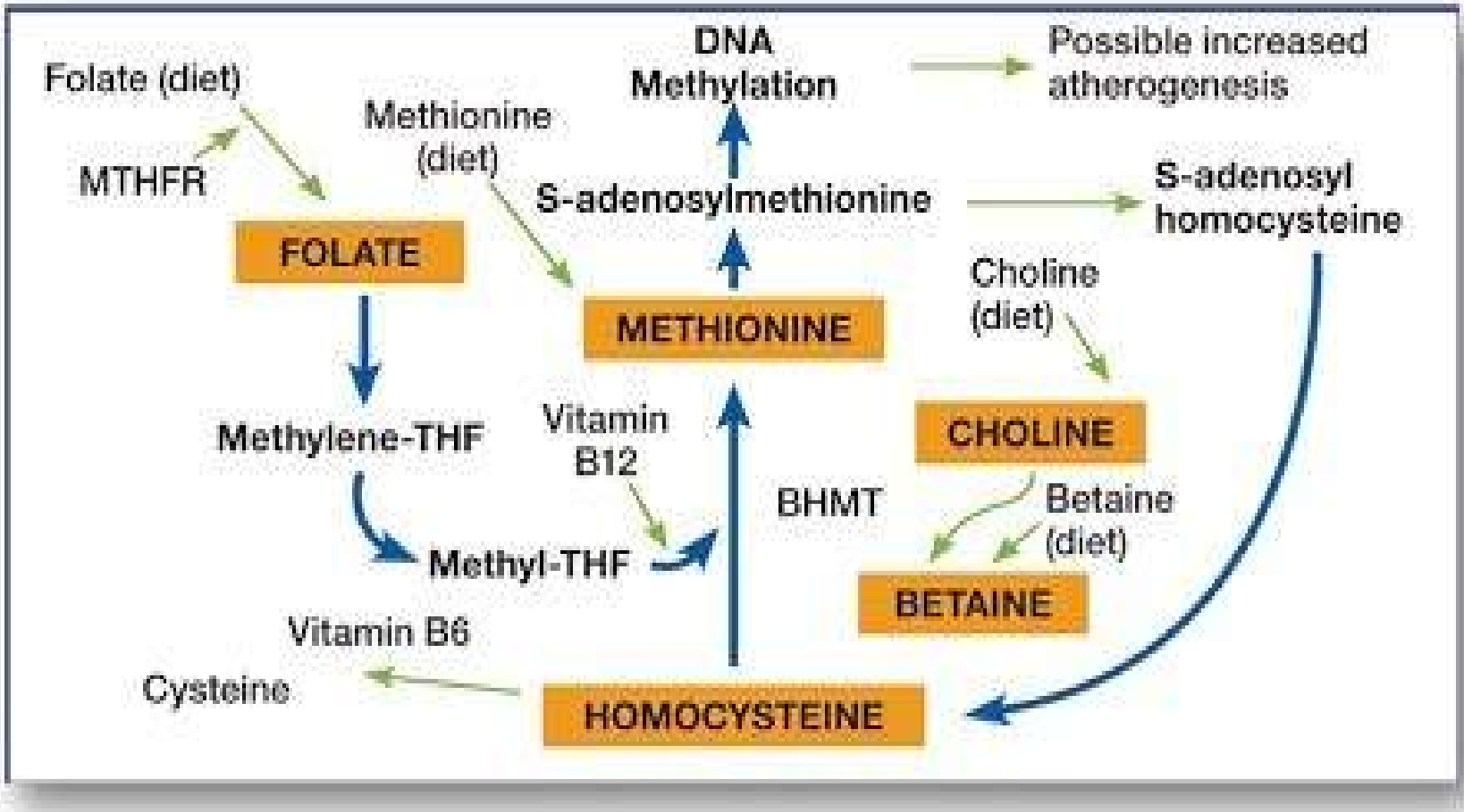
The methyl group of Methionine can be transferred after activation to an acceptor for the formation of important biological compounds.

For example (figure) conversion of norepinephrine to epinephrine by S Adenosyl methionine (active form of methionine



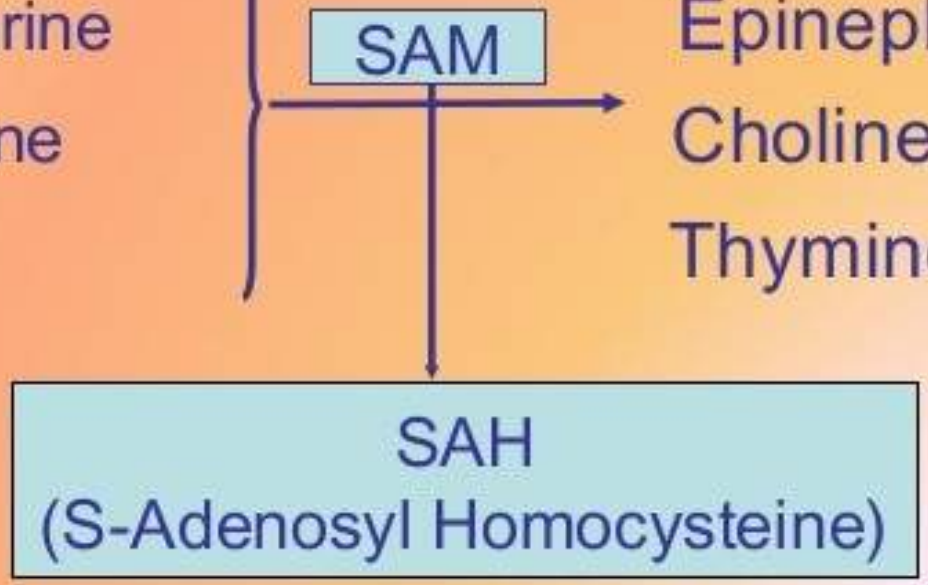


Transmethylation

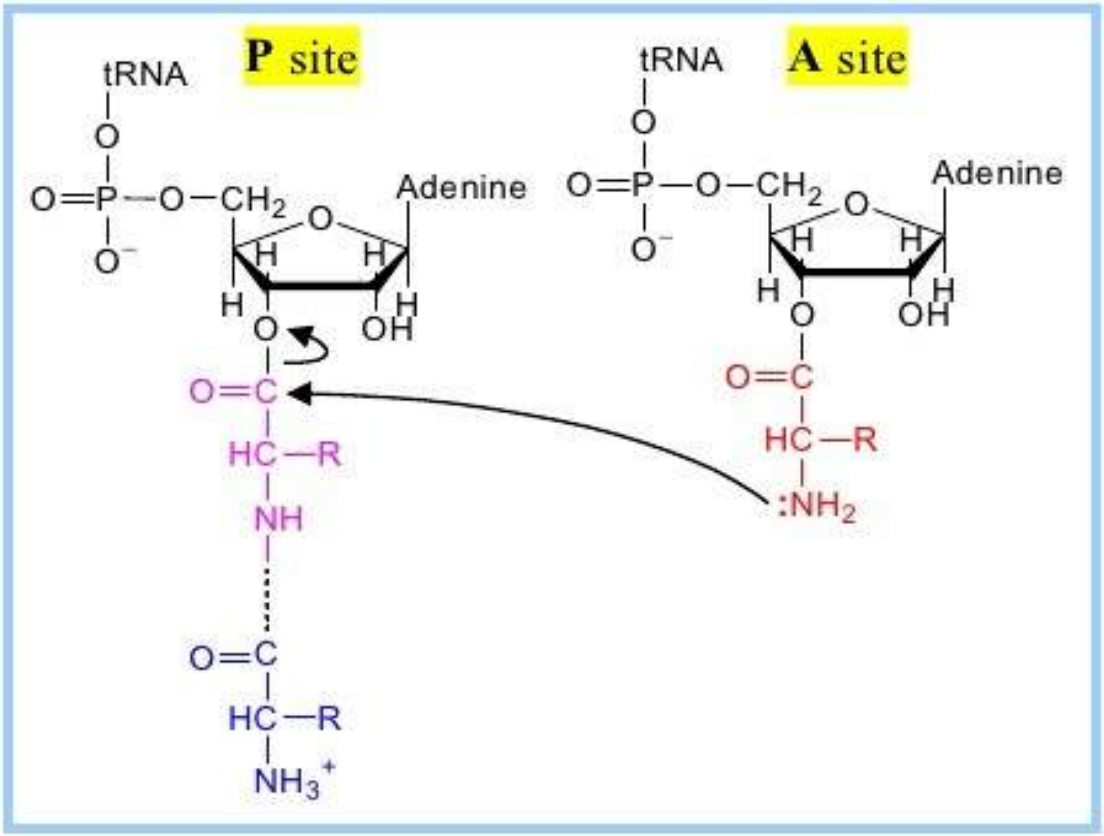


In transmethylation there are:

Methyl acceptors	***Methyl Compounds***
1- Guanidoacetic acid	Creatine
2- Norepinephrine	Epinephrine
3- Ethanolamine	Choline
4- Uracil	Thymine



Transpeptidation (peptide bond formation) involves **nucleophilic attack** of the **amino N** of the amino acid linked to the 2' or 3' OH of the terminal adenosine of the tRNA in the **A site** on the **carbonyl C** of the amino acid (with attached nascent polypeptide) in ester linkage to the tRNA in the **P site**.

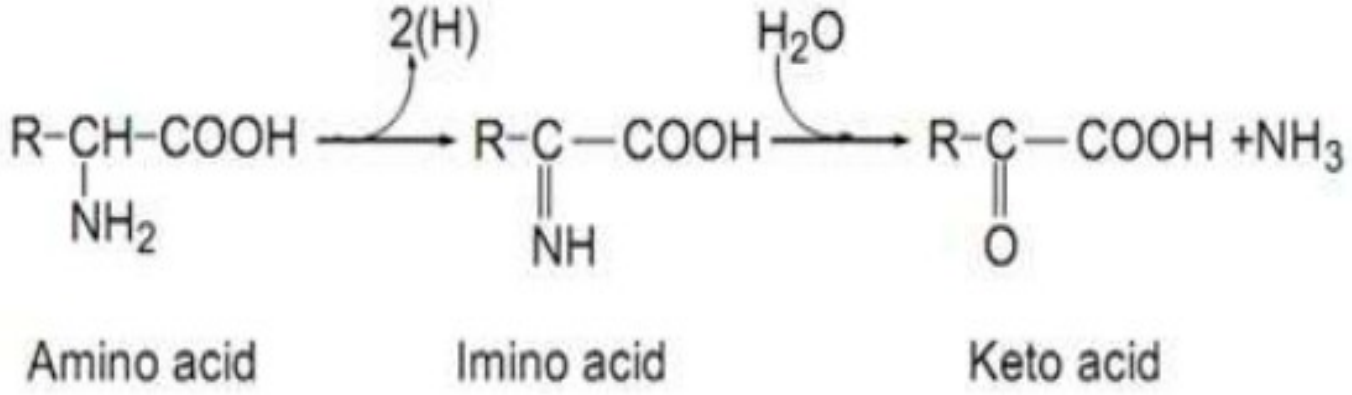


The reaction is promoted by the geometry of the active site consisting solely of residues of the **23S rRNA** of the large ribosomal subunit. **No protein** is found at the active site.

Reactions due to amino group

i) Oxidative deamination-An amino group is removed and corresponding α-keto acid is formed.

α -keto acid produced is either converted to glucose or ketone bodies or is completely oxidized.



Oxidative Deamination

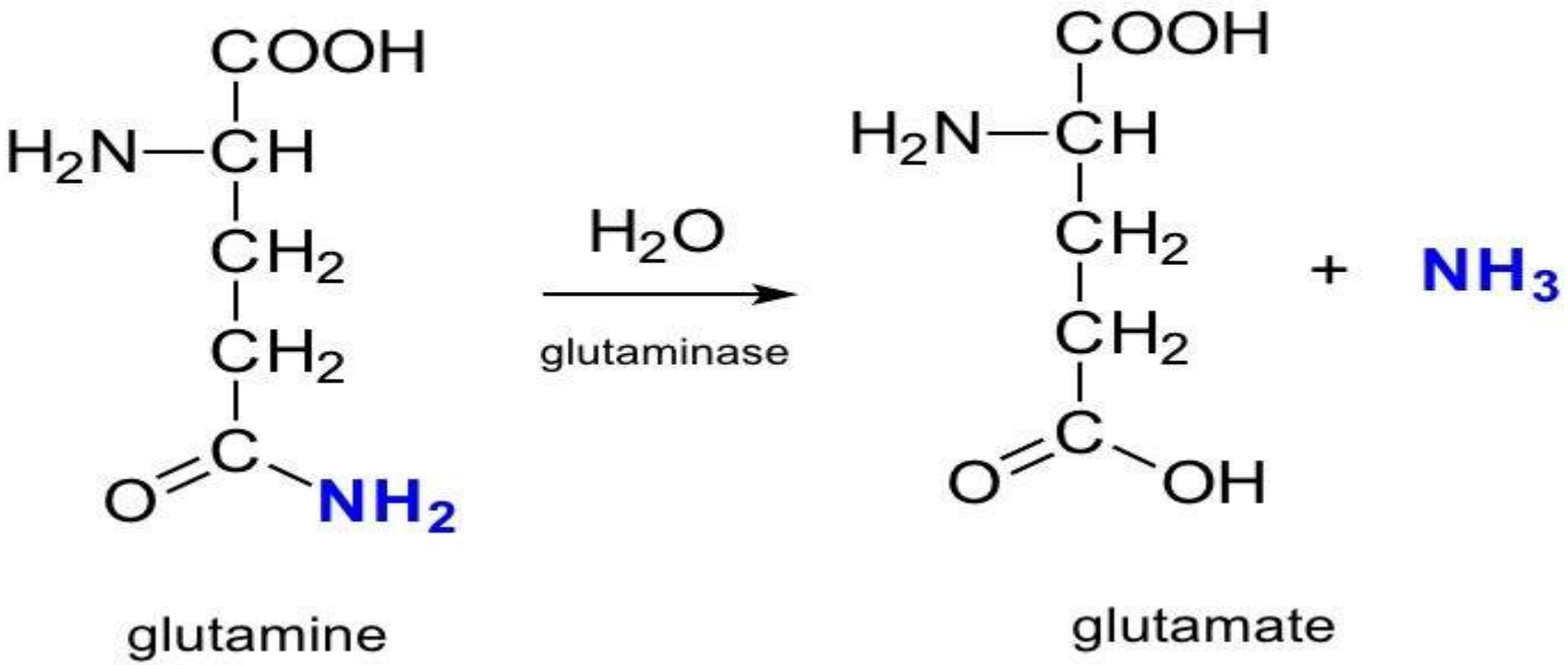


Oxidative Deamination

- **Oxidative deamination** is the liberation of free ammonia from the amino group of amino acids coupled with oxidation.
- **Site**: Mostly in liver & kidney.
- Oxidative deamination is to provide NH_3 for urea synthesis & α-keto acids for a variety of reactions, including energy generation.

Deamidation

Glutaminase is amidase which catalyzes the deamidation of glutamine

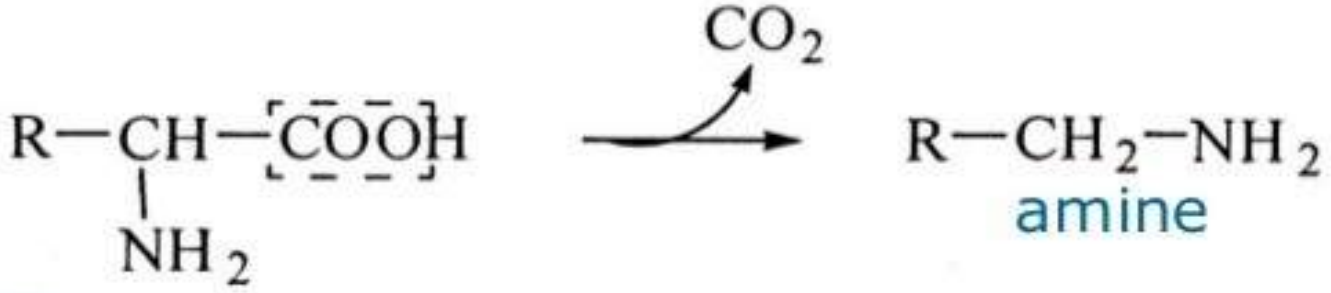


glutamine

glutamate

Decarboxylation of amino acids

Decarboxylation – removal of *carbon dioxide* from amino acid with formation of *amines*.



Usually amines have high physiological activity (hormones, neurotransmitters etc).

Enzyme: ***decarboxylases***

Coenzyme – **pyrydoxalphosphate**

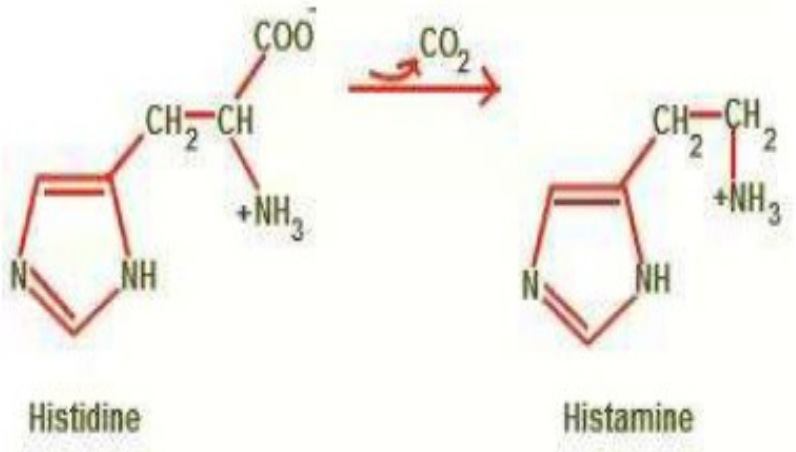
2) Reactions due to carboxyl group

i) Decarboxylation- Amino acids undergo alpha decarboxylation to form corresponding amines.
Examples-

Glutamic acid → GABA

Histidine → Histamine

Tyrosine → Tyramine

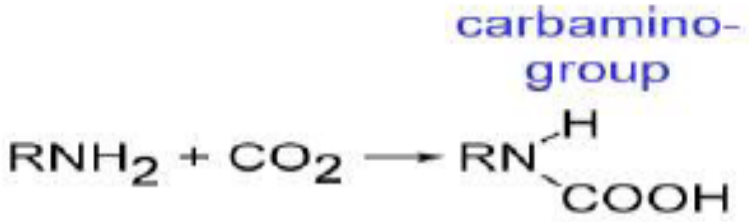


Histidine Histamine

Reactions due to amino group (contd.)

iii) Formation of carbamino compound

- CO_2 binds to α amino acid on the globin chain of hemoglobin to form carbamino hemoglobin
- The reaction takes place at alkaline pH and serves as a mechanism for the transfer of Carbon dioxide from the tissues to the lungs by hemoglobin.



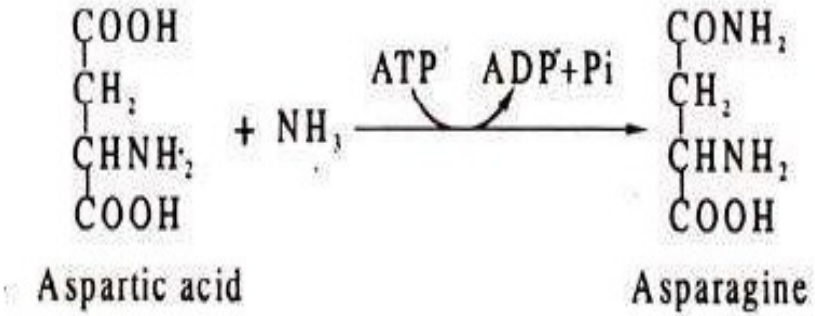
2) Reactions due to carboxyl group (contd.)

ii) Formation of amide linkage

Non α carboxyl group of an acidic amino acid reacts with ammonia by condensation reaction to form corresponding amides

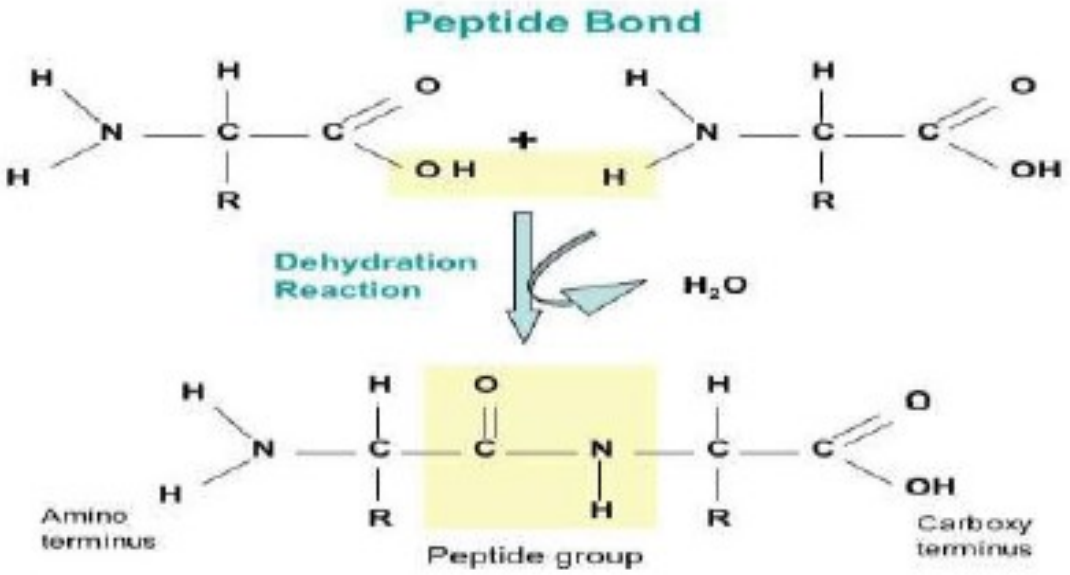
Aspartic acid ⟶ Asparagine

Glutamic acid ⟶ Glutamine



3)Reactions due to both amino & carboxyl groups

Formation of peptide bond- Carboxyl group of an amino acid binds with amino group of another amino acid forming a peptide bond with the loss of one molecule of water.



Reactions due to side chains

1) Ester formation

- OH containing amino acids e.g. serine, threonine can form esters with phosphoric acid in the formation of phosphoproteins (figure-1)
- OH group containing amino acid can also form: Glycosides – by forming O- glycosidic bond with carbohydrate residues (figure-2)

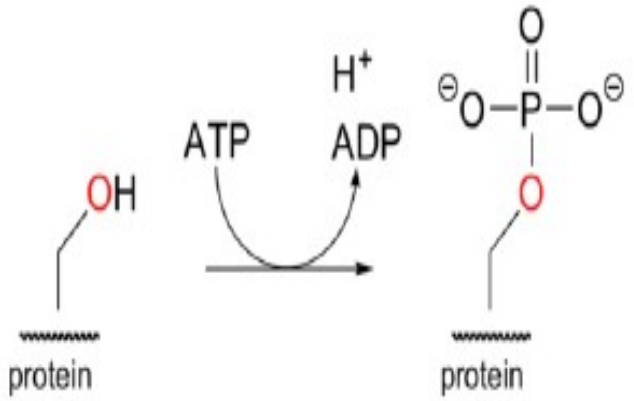


Figure-1

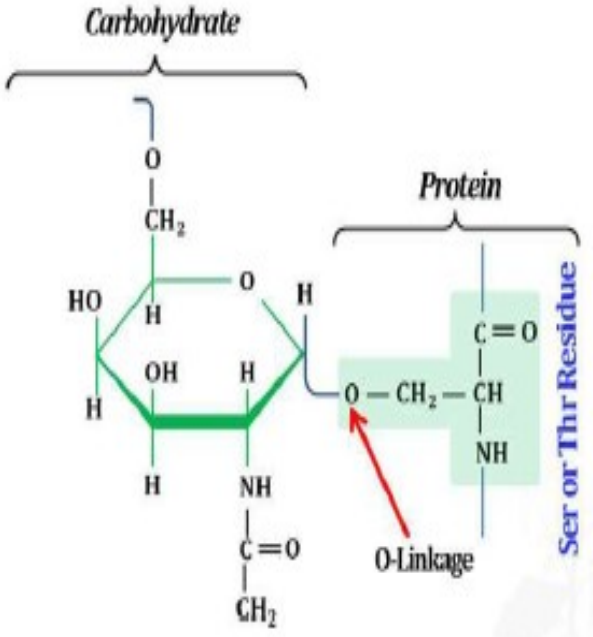
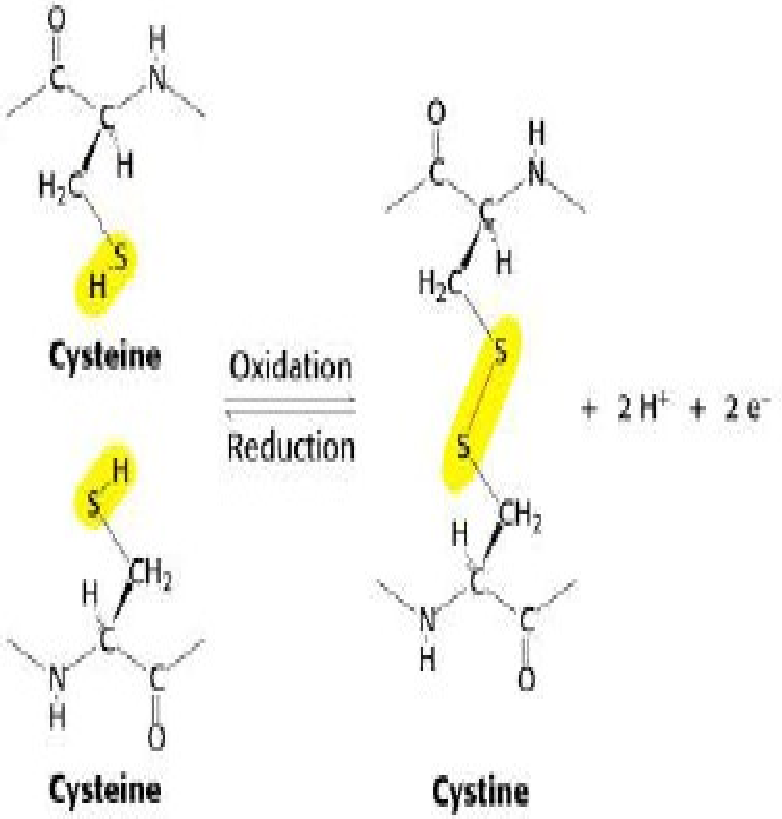


Figure-2

Reactions due to side chains

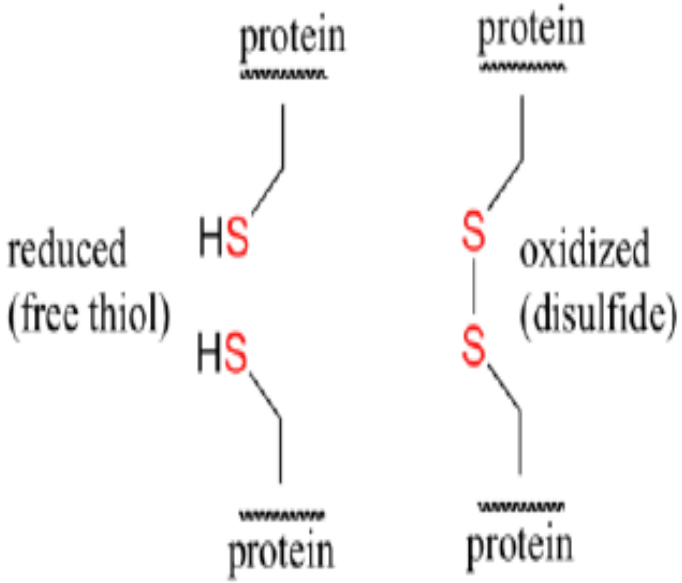
2) Reactions due to SH group (Formation of sulphide bonds)

- Cysteine has a sulfhydryl group(SH) group and can form a disulphide (S-S) bond with another cysteine residue.
- The dimer is called Cystine



Formation of disulphide bond

- Two cysteine residues can connect two polypeptide chains by the formation of interchain disulphide chains.



Thank You